

Ramshri Mohapatra

Data Analyst

London, UK | +44 7833 889303 | rskissan1729@gmail.com
github.com/Ramshri-Mohapatra | linkedin.com/in/ramshri-mohapatra

PROFESSIONAL SUMMARY

First-class Computer Science graduate from City, University of London, with hands-on experience in data pipeline engineering, database design, and applied statistical and ML methods for drawing conclusions from real-world, noisy data. Built ETL pipelines processing live datasets, deployed multiple production systems independently, and gained hands-on exposure to BI dashboarding and CRM data during a consulting internship. Right to work in the UK.

TECHNICAL SKILLS

Languages: SQL, Python, C++, Java

BI & Reporting: Excel, Power BI, Tableau, Plotly, Streamlit, Matplotlib

Data & Storage: PostgreSQL, SQLAlchemy ORM, MongoDB, data modelling, ETL pipelines

Data & ML: pandas, NumPy, scikit-learn, HuggingFace Transformers, NLP, TF-IDF, GridSearchCV, statistical analysis, data validation

Backend: FastAPI, RESTful API design, JWT auth, OAuth2, dependency injection, Pydantic

Cloud & Infra: Docker, docker-compose, GitHub Actions CI/CD, Google Cloud VM, Railway, Render, Linux, Git

Testing: pytest, test isolation, fixtures, security & authorization testing

PROJECTS

Fintech Backend API

2026 | [Live](#) + [GitHub](#)

- Production-grade personal finance REST API with JWT auth (bcrypt, python-jose) and full CRUD with strict per-user data isolation enforced at the query level.
- Relational schema in SQLAlchemy over PostgreSQL; 12 pytest tests against an isolated SQLite DB covering auth, authorization failures and cross-user isolation.
- Containerised with Docker; GitHub Actions CI runs the full suite on every push; deployed with environment-based secrets — initially on Railway, then redeployed to Render.

Algorithmic Trading Backtester

2026 | [Live](#) + [GitHub](#)

- End-to-end trading system: pipeline fetching real OHLCV data with retry logic, indicators from scratch (SMA, EMA, RSI), two strategies backtested with look-ahead bias prevention.
- RSI outperformed MA Crossover on MSFT (Sharpe 0.73 vs 0.50, drawdown -28% vs -39%). Deployed a live Streamlit + Plotly dashboard.

VIGIL-AI: Cloud Data Pipeline — Dissertation (80%, First Class)

2024–2025 | [GitHub](#)

- Long-running ETL pipeline on Google Cloud VM: automated ingestion, transformation, deduplication and MongoDB storage with schema validation and audit logging.
- Fine-tuned DeBERTa v3 for NER on cybersecurity text, F1 66.3% after systematic hyperparameter tuning. Accepted at ICETCS and ITSS-IoE 2025; pending Springer Nature publication.

Stock Sentiment Analysis

2025–2026 | [GitHub](#)

- Compared three NLP classifiers (BoW + Random Forest, TF-IDF + sentiment + tuned Logistic Regression, LR/RF/SVM voting ensemble) predicting stock direction from 4,101 days of headlines, with GridSearchCV tuning and chronological splits.
- Found and fixed a silent split bug: an unparsed date column turned a date filter into a string comparison, shrinking the test set 378 to 126 days and inflating accuracy to 56.3% against a corrected 51.6%.
- Ruled out a broken pipeline via a positive control (81.3% on a corpus with known sentiment signal) and a permutation test (the original 56.3% claim was within noise, $p = 0.14$).

Email Spam Classification

2026 | [GitHub](#)

- Built a spam-detection pipeline over 33.7k labelled Enron emails (TF-IDF into a decision tree, linear SVM, PyTorch MLP); best model reached 98.7% accuracy, 0.987 F1, 0.999 ROC-AUC.
- Audited my group's original coursework and found its labels were keyword-matched from the same text fed to the model; replaced the corpus and fixed three independent sources of train/test leakage.

- Validated against an independent human-labelled mailbox: recall fell from 0.99 to 0.64 on real mail, every miss a newsletter or stock alert absent from the training spam.

EXPERIENCE

Bartender (Part-Time), Stonegate Group *Aug 2024 – Present*

Student Ambassador, Unitemps *Dec 2022 – Apr 2025*

- Delivered talks, workshops and one-to-one guidance across 50+ higher-education fairs (audiences of 50–200), translating university processes into accessible guidance for non-specialists.

Unity 3D Developer Intern, Constituents AI and Technology *Apr – Sep 2024*

- Led end-to-end delivery of client-facing VR education packages in a structured GitHub workflow; progressed to leading a team of 3 interns within 3 months.
- Reduced load times by 40% through data-driven profiling; communicated findings to non-technical stakeholders.

Summer Intern, GoldenPegasus IT Consulting *Jun – Sep 2023*

- Built Power BI dashboards and used Excel for ad hoc analysis of Salesforce CRM data in a client-facing consulting environment.

EDUCATION

BSc (Hons) Computer Science, First Class — City, University of London *2022–2025*

- Dissertation: VIGIL-AI (80%). Accepted at ICETCS and ITSS-IoE 2025, pending Springer Nature publication.
- President's International Scholarship Awardee (2022–2023).
- Key modules: Algorithms & Data Structures, Distributed Systems, Statistics, Databases (SQL), AI, NLP, Software Engineering.